Program - 6

Exploratory Data Analysis

Date : 12.09.2026

Aim

Carry out Exploratory Data Analysis (EDA) using Python by generating statistical summaries and visualizations such as histograms, boxplots, and heatmaps.

Algorithm

1. Start
2. Import the pandas and numpy libraries.
3. Create a dictionary containing student details such as Student ID, Name, Marks, and Attendance.
4. Convert the dictionary into a Pandas DataFrame named df.
5. Display the original dataset.
6. Check and display the number of missing values in each column.
7. Replace missing values in the Marks column with the mean of the available Marks.
8. Replace missing values in the Attendance column with the mean of the available Attendance values.
9. Display the dataset after handling missing values.
10. Count and display the number of duplicate records.
11. Remove duplicate rows using drop_duplicates().
12. Display the dataset after removing duplicates.
13. Apply Min-Max Normalization to the Marks column.
14. Apply Min-Max Normalization to the Attendance column.
15. Display the normalized dataset.
16. Calculate Q1 (First Quartile) and Q3 (Third Quartile) for Marks.
17 Calculate the Interquartile Range (IQR): IQR = Q3 - Q1
18. Calculate the lower limit and upper limit for outlier detection.
19. Identify and display Marks values that fall outside these limits as outliers.
20. Remove the detected outliers from the dataset.
21. Store the resulting dataset as df_clean.
22. Display the final preprocessed dataset.
23. Save the final dataset as preprocessed_students.csv.
24. Display a message confirming that the preprocessed dataset has been saved successfully.
25. Stop.

Program - 6

```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
sns.set_theme(style="ticks")
plt.rcParams["font.size"] = 10
df = sns.load_dataset("iris")
print("--- 1. FIRST 5 ROWS OF DATA ---")
print(df.head(), "\n")
print("--- 2. DATASET STRUCTURE & TYPES ---")
print(df.info(), "\n")
print("--- 3. MISSING VALUE COUNTS ---")
print(df.isnull().sum(), "\n")
print("--- 4. DESCRIPTIVE STATISTICAL METRICS ---")
print(df.describe(), "\n")
print("--- 5. CATEGORICAL VALUE COUNTS ---")
print(df["species"].value_counts(), "\n")
fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(18, 5))
fig.suptitle(
    "Exploratory Data Analysis (EDA) Dashboard",
    fontsize=15,
    fontweight="bold",
    y=1.02,
)
sns.histplot(
    data=df,
    x="sepal_length",
    hue="species",
    kde=True,
    multiple="stack",
    palette="muted",
    ax=axes[0],
)
axes[0].set_title("A. Distribution of Sepal Length")
axes[0].set_xlabel("Sepal Length (cm)")
axes[0].set_ylabel("Frequency Count")
sns.boxplot(data=df, x="species", y="petal_width", palette="pastel", ax=axes[1])
axes[1].set_title("B. Petal Width Across Species")
axes[1].set_xlabel("Flower Species")
axes[1].set_ylabel("Petal Width (cm)")
correlation_matrix = df.select_dtypes(include="number").corr()
sns.heatmap(
    data=correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
```

```
        linewidths=0.5,
        vmin=-1,
        vmax=1,
        ax=axes[2],
)
axes[2].set_title("C. Feature Correlation Matrix")
plt.tight_layout()
plt.show()
```